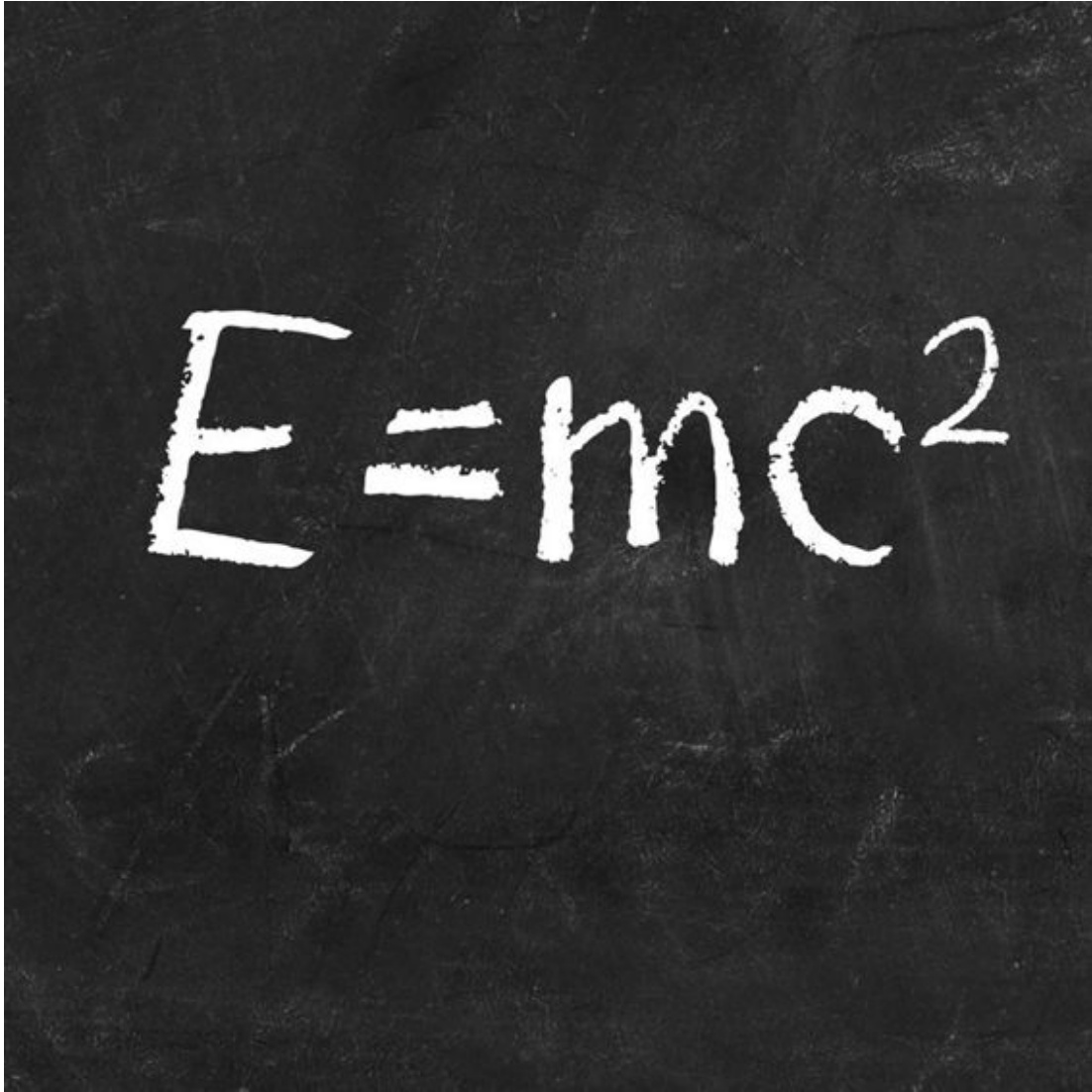


Scientific answers to metaphysical questions

A black and white photograph of a chalkboard with the equation $E=mc^2$ written in white chalk. The chalkboard has a dark, textured surface with some faint, illegible markings in the background.

Einstein's famous equation shows that energy (***E***) is equal to the product of mass (***m***) and the square of the speed of light (***c***).

When Einstein formulated his theories of relativity at the beginning of the 20th century, he ushered in a completely new way of understanding the universe. This overturned the old Newtonian view, replacing it with a comprehensive description of the physical universe according to a new set of physical laws. For the first time, time and space themselves were explained scientifically, and the substance of the universe was defined in

terms of energy. Many of the metaphysical questions that had concerned philosophy from its earliest beginnings seemed finally to be answered. But, as Einstein would have happily admitted, the new physics was not a definitive answer, nor did it negate the importance of Newton's contribution. It was not 'right' or 'true', but simply a more accurate explanation than Newton's, which was perfectly good for its time – as a pragmatist would say, it was a valid explanation. And just as Newtonian physics marked a stage in the history of science, so too Einstein's theories might one day be replaced with something that fits the facts even better.